Project Proposal

Luxury Pulse: Global Fashion Retail Performance Dashboard

Written by: Roshni Patel (20036005)

Project Overview

Track Selection I am pursuing Track 1: Executive Dashboard.

Project Title & Description: My project is called Luxury Pulse an interactive, multi-page executive dashboard that analyzes performance across the global fashion and luxury retail industry. Using publicly available retail transaction data, luxury product catalogs, and brand search trend data, the dashboard gives business users a comprehensive view of sales performance, product pricing tiers, geographic distribution, and brand awareness all in one place.

Motivation: Fashion retail is one of the most data-rich consumer industries, yet most publicly available analyses only scratch the surface. I chose this domain because it sits at the intersection of business strategy, consumer behavior, and brand perception all of which are highly visual and compelling to analyze. I also want to build something I can speak about confidently in data analyst job interviews, and a fashion retail dashboard is specific enough to be memorable while being broad enough to demonstrate a wide range of analytical skills.

Goals & Objectives:

- Build a fully interactive Tableau dashboard with 8+ linked visualizations across multiple pages.
- Analyze sales performance, product pricing, geographic trends, and brand awareness in one unified view.
- Surface at least 5 actionable business insights from the data.
- Publish the final dashboard on Tableau Public with a shareable link for my resume and portfolio.
- Document the full process including data cleaning, design decisions, and AI-assisted workflows.

Target Audience: My primary audience is fashion retail business analysts and brand managers who need to monitor performance across categories and markets. My secondary audience is hiring managers reviewing my portfolio, for whom this project demonstrates end-to-end data analyst skills in a visually strong domain.

Data Plan

Data Sources: I will use four free Kaggle datasets combined with one free Python library:

- **Global Fashion Retail Stores Dataset** This is my core dataset. It contains two years of multinational fashion brand sales simulation covering transactions, product categories, store locations, and geographic breakdowns across multiple countries. The entire dashboard spine is built around this.

- **Net-a-Porter / Mr Porter Luxury Fashion Dataset** Real product listings from two of the world's leading luxury e-commerce platforms, including brand names, categories, price points, and availability. I will use this to build the product pricing and luxury tier analysis page.
- **Fashion Retail Sales Dataset** Transaction-level data with order values, customer review scores, and payment methods. I will use this to power KPI cards around average order value and customer satisfaction metrics.
- **Luxury Apparel Data** Detailed luxury clothing product records with price tiers and category labels. Useful for premium vs. entry-level segment breakdowns to complement the Net-a-Porter data.
- **pytrends** (free Python library, pip install, no API key) I will pull Google Trends search interest data for major fashion brand keywords by country to build a brand awareness and market heat page.

Data Description

Dataset	Format	Est. Size	Key Variables
Global Fashion Retail Stores	CSV	500K+ rows	Date, Store, Country, Category, Revenue, Units Sold
Net-a-Porter / Mr Porter	CSV	~10K rows	Brand, Category, Price, Currency, Availability
Fashion Retail Sales	CSV	~3K rows	Order Value, Review Score, Payment Type, Date
Luxury Apparel Data	CSV	~2K rows	Brand, Price Tier, Category, Description
Google Trends (pytrends)	DataFrame → CSV	~5K rows	Date, Brand Keyword, Country, Interest Score (0–100)

Access Verification All four Kaggle datasets are freely downloadable with a Kaggle account no special permissions, no approval process. I have confirmed all four are currently live and accessible. pytrends is an open-source Python library with no API key required.

Anticipated Challenges The biggest challenge will be schema alignment each CSV uses different column naming conventions, so I will write a standardized Python cleaning script to unify them before loading into Tableau. Currency is another issue since the Net-a-Porter dataset lists prices in GBP; I will convert to USD using a fixed average exchange rate and disclose this in the dashboard. For Google Trends, scores are relative (0–100 per query) rather than absolute search volumes, which I will document clearly in tooltip text, so users don't misread the chart. Finally, the primary Global Fashion Retail dataset is a realistic sales simulation rather than live transactional data. I will be transparent about this in the dashboard's data source footnote.

Preparation Plan

1. Download all four Kaggle CSVs → inspect schemas, null rates, and value distributions using pandas
2. Clean and standardize each dataset individually rename columns, fix data types, handle nulls, convert GBP to USD
3. Pull Google Trends data via pytrends for 8–10 major fashion brand keywords → export to CSV
4. Join all five sources into a single master Tableau-ready dataset
5. Validate with summary statistics and spot-check key figures before importing to Tableau

Technical Plan

Tools & Technologies

- **Python** (pandas, pytrends) — data acquisition, cleaning, transformation, and export
- **Jupyter Notebook** — exploratory analysis and pipeline documentation
- **Tableau Public** — primary dashboard platform; free tier supports public link sharing which makes it immediately portfolio-ready
- **Claude AI** — used throughout the project for code generation, Tableau calculated field help, design feedback, and documentation

Visualization Plan & Dashboard Structure

I am building four dashboard pages, each focused on a distinct business question:

Page 1: Sales Overview *Question: How is the business performing overall and what's driving revenue?*

- Revenue trend line over time (monthly)
- Top 10 product categories by revenue (horizontal bar chart)
- Year-over-year growth rate KPI scorecards
- Revenue by store count heatmap

Page 2: Geographic Performance *Question: Which markets are strongest and where is growth happening?*

- Filled world map showing revenue by country
- Regional revenue breakdown (stacked bar)
- Top 5 countries by average order value (ranked bar)
- Country-level growth rate comparison

Page 3: Product & Brand Intelligence *Question: How is the product mix structured across price tiers?*

- Price tier distribution (histogram)

- Luxury vs. mid-market segment comparison (side-by-side bars)
- Average price by category (dot plot)
- Product availability rate by brand (lollipop chart)

Page 4: Brand Pulse *Question: Which brands have the most consumer mindshare and where?*

- Google Trends line chart for top brand keywords over time
- Brand search interest by country (bar chart)
- Seasonal search interest patterns (heat calendar)
- Review score distribution by payment type (box plot)

Interaction Design

- Global category filter propagating across all four pages
- Date range slider controlling all time-series charts
- Hover tooltips with full data context on every mark
- Drill-down on the geographic map to country-level detail
- Parameter toggle to switch between revenue, units sold, and average order value on key charts
- Mobile-responsive layout using Tableau's device preview

AI Strategy: I will use Claude AI at every stage. During data prep, I'll use it to generate and debug Python cleaning scripts. During Tableau development, I'll use it to write complex calculated fields especially LOD expressions for YoY growth rates, running totals, and country-level ranking. For design decisions, I'll use it as a sounding board for chart type selection and color palette choices. I'll also use it to help write the user guide and generate talking points for my final presentation.

Technical Risks & Mitigations

- Tableau Public performance with 500K+ rows → I will pre-aggregate the core dataset in Python to summary-level tables before import, keeping Tableau extracts under 50K rows
- pytrends rate limiting → I will pull brand data in batches with time delays between requests and cache results to CSV immediately
- Small sample sizes in Fashion Retail Sales (~3K rows) → I will use this dataset only for KPI cards and review analysis, not as a primary source for trend charts

Timeline & Milestones

Phase	Key Tasks	Deliverable
Week 10	• Submit proposal • Download all 4 Kaggle datasets	Proposal doc + EDA notebook

	• Run EDA in Jupyter • Pull Google Trends via pytrends	
Week 11	• Complete Python cleaning pipeline • Build first 3–4 Tableau charts • Confirm all data joins work • Progress check-in	Clean master dataset + draft dashboard skeleton
Week 12	• Build all 8+ visualizations • Add all filters and interactions • Mobile layout • Peer review session	Full interactive dashboard
Week 13	• User testing with 5 testers • Incorporate feedback • Write user guide • Record demo walkthrough • Publish to Tableau Public	Final portfolio deliverable + shareable link